

# Shivang Chopra

Ph.D. in Computer Science at Georgia Institute of Technology

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Google Scholar: [🔗](#)

## RESEARCH INTERESTS

**Robust Representation Learning:** Investigating the algorithmic and geometric principles underlying generalization in multimodal foundation models, with a focus on representation structure, robustness, and efficient adaptation.

**Spatial Reasoning and World Models:** Developing methods to integrate structured latent representations and inductive biases from world models into vision-language systems, enabling enhanced spatial reasoning in Vision Language Models.

**Test-Time Scaling and Adaptive Computation:** Designing inference-time scaling mechanisms through adaptive computation and virtual depth, enabling pretrained models to achieve stronger reasoning without increasing parameter count.

## EDUCATION

### Georgia Institute of Technology

Jan 2025 - May 2029 (expected)

*Ph.D. in Computer Science*

Advisor: Prof. Zsolt Kira [🔗](#)

Publications: CVPR 2026 [2], CVPR(W) 2025 [8], ICLR 2025 [4]

### Georgia Institute of Technology

Aug 2022 - Dec 2024

*Master of Science in Computer Science*

**CGPA:** 3.9 / 4.0

Thesis: CRAFT: Curriculum Rank Adversarial Fine-Tuning for Robust Vision Language Models [📄](#)

Publications: WACV 2025 [3], CoRL 2023 [5], ICML(W) 2022 [10]

### Delhi Technological University

Aug 2016 - May 2020

*Bachelor of Technology in Computer Engineering*

**CGPA:** 8.6 / 10.0

Publications: AAAI 2020 [7], IEEE CDS [6]

## RESEARCH EXPERIENCE

### Sony R&D Labs

Zurich, Switzerland

*Computer Vision Research Intern*

Sep 2023 - Apr 2024

**Research Focus:** Inductive biases and robustness in object tracking under extreme distribution shift

- Identified conditions under which asynchronous temporal sparsity induces implicit regularization in event-based vision models
- Demonstrated improved tracking stability across lighting, motion, and noise variations
- Proposed event-density-aware representation learning framework improving robustness of event-based object tracking under extreme distribution shifts.

### Amazon Science

Washington, USA

*Applied Scientist Intern*

May 2023 - Aug 2023

**Research Focus:** Source-Free Domain Adaptation (SFDA) using generative priors [9]

- Studied pretrained diffusion models as generative priors for source-free domain adaptation, enabling recovery of source-like representations using only target-domain data.
- Proposed a diffusion-guided sample generation and mixup strategy that bridges source-target gaps by generating domain-aligned images, yielding improved generalization across standard SFDA benchmarks.

### Microsoft Research Lab India

Bangalore, India

*Research Intern*

May 2022 - Aug 2022

**Research Focus:** Multimodal perception for accessibility and social presence.

- Designed a multimodal perception subsystem integrating vision, audio, and gaze cues to support informal interaction in hybrid physical-virtual environments.
- Studied how spatial and directional cues affect accessibility and social awareness for remote participants.
- *U.S. patent granted for the AI prototype [P1]*

### IBM Research Lab

Delhi, India

*Research Intern*

June 2020 - Aug 2020

**Project:** Multimodal content understanding with user-driven personalization ([Project Report](#))

- Developed a multimodal framework for explicit content detection that preserves narrative and semantic continuity rather than frame-level filtering.
- Explored real-time deployment constraints by integrating model outputs into a user-facing browser extension.

## PUBLICATIONS

### Under Review

- [1] **Thinking Deeper Without Growing Larger: Scaling Virtual Depth in Pretrained LLMs for Enhanced Reasoning**

Shivang Chopra, Zhang Xi-Jia, Karmesh Yadav, Yusuf Ali, Chengyue Huang, Zsolt Kira.

### Conferences and Journals

- [2] **The Geometry of Robustness: Optimizing Loss Landscape Curvature and Feature Manifold Alignment for Robust Finetuning of Vision-Language Models**

Shivang Chopra, Shaunak Halbe, Chengyue Huang, Brisa Maneechotesuwan, Zsolt Kira.

Paper accepted at IEEE/CVF Conference on computer vision and pattern recognition, [CVPR 2026](#)

[3] **Refining Text-to-Image Generation: Towards Accurate Training-Free Glyph-Enhanced Image Generation**  
Sanyam Lakhanpal, **Shivang Chopra**, Vinija Jain, Aman Chadha, Man Luo  
Published at IEEE/CVF Winter Conference on Applications of Computer Vision, [WACV 2025](#)

[4] **Directional Gradient Projection for Robust Fine-tuning of Foundation Models**  
Chengyue Huang, Junjiao Tian, Brisa Maneechotesuwan, **Shivang Chopra**, Zsolt Kira  
Published at the International Conference on Learning Representations, [ICLR 2025](#)

[5] **Learning to Discern: Imitating Heterogeneous Human Demonstrations with Preference and Representation Learning**  
Sachit Kuhar, Shuo Cheng, **Shivang Chopra**, Matthew Bronars, Danfei Xu  
Published at the Conference on Robot Learning, [CoRL 2023](#)

[6] **Attention-Net: An Ensemble Sketch Recognition Approach using Vector Image**  
**Shivang Chopra\***, Gaurav Jain\*, Suransh Chopra\*, Anil Singh Parihar  
Published in IEEE Transactions on Cognitive and Developmental Systems, [IEEE CDS 2022](#)

[7] **Hindi-English Hate Speech Detection: Author Profiling, Debiasing, and Practical Perspectives**  
**Shivang Chopra**, Ramit Sawhney, Puneet Mathur, Rajiv Ratn Shah  
Published at the AAAI Conference on Artificial Intelligence, [AAAI 2020](#)

Workshops/Preprint

[8] **MedMoE: Modality-Specialized Mixture of Experts for Medical Vision-Language Understanding**  
**Shivang Chopra**, Lingchao Mao, Gabriela Sanchez-Rodriguez, Andrew J. Feola, Jing Li, Zsolt Kira  
Published at the [MMFF-BIOMED workshop at CVPR 2025](#).

[9] **Source-free domain adaptation with diffusion-guided source data generation**  
**Shivang Chopra**, Suraj Kothawade, Houda Aynaou, Aman Chadha. [arXiv 2024](#)

[10] **Active Data Discovery: Mining Unknown Data using Submodular Information Measures**  
Suraj Kothawade, **Shivang Chopra**, Saikat Ghosh, Rishabh Iyer  
Published at the [RealML workshop at ICML 2022](#).

Patent

[P1] Hybrid Environment for Interactions Between Virtual and Physical Users (**US 18/358,485**), 2024

AWARDS AND ACHIEVEMENTS

|                                                                                                 |           |
|-------------------------------------------------------------------------------------------------|-----------|
| <b>Graduate Student Scholarship</b> , Georgia Institute of Technology                           | 2023,2024 |
| <b>Research Excellence Award</b> , Delhi Technological University                               | 2020      |
| <b>Student Research Travel Grant</b> , Microsoft Research India                                 | 2020      |
| <b>Qualified for Onsite Regionals</b> , ACM ICPC (International Collegiate Programming Contest) | 2018      |

RELEVANT COURSEWORK

|                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ph.D. / Master Of Science:</b> Machine Learning, Vision Language Models, Deep Learning for Robotics, Game AI, Natural Language Processing, Learning and Memory, Cognitive Psychology                                                    |
| <b>Bachelor of Technology:</b> Machine Learning, Artificial Intelligence, Natural Language Processing, Distributed Systems, Operating Systems, Computer Networks, Information and Network Security, Computer Organization and Architecture |

TECHNICAL SKILLS

|                                 |                                                         |
|---------------------------------|---------------------------------------------------------|
| <b>Programming Languages</b>    | Python, Bash, C, C++                                    |
| <b>Deep Learning Frameworks</b> | PyTorch, TensorFlow, JAX, Accelerate, Pytorch Lightning |
| <b>Distributed System Tools</b> | Slurm, Docker, Kubernetes                               |

COMMUNITY AND ACADEMIC SERVICES

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| <b>Reviewer:</b> CVPR 2026, NeurIps 2024-25, ICLR 2025, ICML 2025                                                                                                                     |
| <b>Student Volunteer:</b> AAAI 2020, ICRA 2025                                                                                                                                        |
| <b>Project Lead, AI@GT (Georgia Tech):</b> Mentoring a group of three undergraduate students to build self-evolving embodied agents that learn by interacting with their environment. |
| <b>Admin, GT Skynet:</b> Part of the team maintaining Georgia Tech’s SLURM-based GPU cluster to support ML research and coursework.                                                   |

REFERENCES

|                                                                                                |                                                                              |                                                                                                   |                                                                                  |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Prof. Zsolt Kira<br>Associate Professor<br>Georgia Institute of Technology<br>zkira@gatech.edu | Dr. Suraj Kothawade<br>Senior ML Engineer<br>Google<br>skothawade@google.com | Dr. Venkat Padmanabhan<br>Managing Director<br>Microsoft Research India<br>padmanab@microsoft.com | Valentina Cavinato<br>Senior Engineer<br>Sony R&D<br>Valentina.Cavinato@sony.com |
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